California Earthquakes Time Series Analysis

An ARIMA and Spectral Approach to Seismic Activity

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Table of contents

Key Findings . . . 1
Abstract . . . 2
1. Introduction & Data Acquisition . . . 2
    Regional Mapping (EDA) . . . 2
2. Time Domain Analysis . . . 4
    STL Decomposition . . . 4
    Autocorrelation (ACF) . . . 5
3. Spectral Domain Analysis . . . 6
4. Modeling & Forecasting . . . 7
Limitations . . . 8
Conclusion . . . 9
Appendix: R Code Repository . . . 9

Key Findings

- hotspot mapping identifies the Mendocino Triple Junction and Salton Sea/Brawley zone as key localized regions
- STL suggests little meaningful seasonality
- ACF supports clustering/autoregressive behavior
- the smoothed periodogram does not support annual seasonality
- ARIMA and ETS forecasts both show mean reversion, with much wider uncertainty for the northern hotspot than the southern one.

Abstract

This report presents a comprehensive time series analysis of earthquake frequencies in California from 2014 to 2024. Utilizing the U.S. Geological Survey (USGS) Earthquake Catalog API, spatial data bounding boxes were constructed to subset the overall California seismic activity into distinct regional hotspots (Northern California Coast and the Salton Sea area). We conduct exploratory spatial mapping, alongside a rigorous comparison of Time Domain modeling (STL Decomposition, Autocorrelation) against Spectral Domain analysis (Smoothed Periodograms). Finally, we specify Auto-Regressive Integrated Moving Average (ARIMA) and structural State-Space models (ETS) to forecast baseline expected earthquake event counts for the following twelve months. The response variable is the monthly number of earthquakes with magnitude 3.0 or greater, modeled separately for all California, the northern hotspot, and the southern hotspot.

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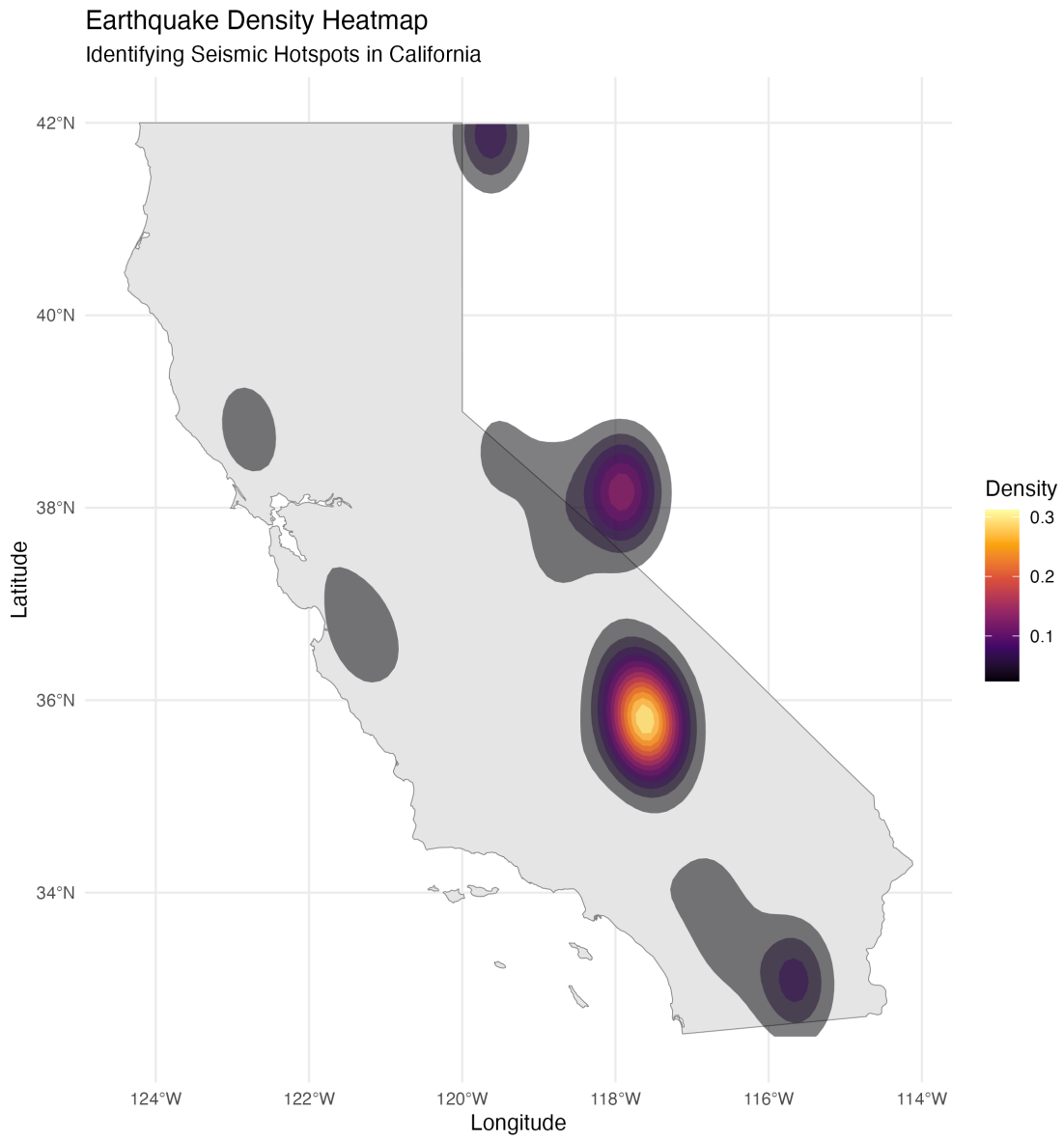
1. Introduction & Data Acquisition

Analyzing earthquake frequencies lies at the intersection of seismology and statistical modeling. An inherent challenge is that tectonic stress is an unobserved phenomenon (the "State"), whereas the earthquakes themselves are the observed realizations of that stress (the "Observation" or "Error").

Using R's `httr2` and `sf` libraries, data was obtained from the USGS for all measured events with a magnitude (M) of 3.0 or greater. The data was aggregated into monthly observation counts to normalize against the varying sensitivity of regional seismographs.

Regional Mapping (EDA)

Before modeling the counts over time, it is vital to model the counts over space. Using Kernel Density Estimation, we identified two localized zones of intense activity to isolate for localized statistical forecasting alongside the aggregate state data.



- **Northern California Hotspot:** The Mendocino Triple Junction.
- **Southern California Hotspot:** The Salton Sea / Brawley Seismic Zone.

These zones were chosen because they both show a dense zone on the KDE heatmap, and are geographically distant enough to be accurate representations of earthquakes in Northern California and Southern California respectively.

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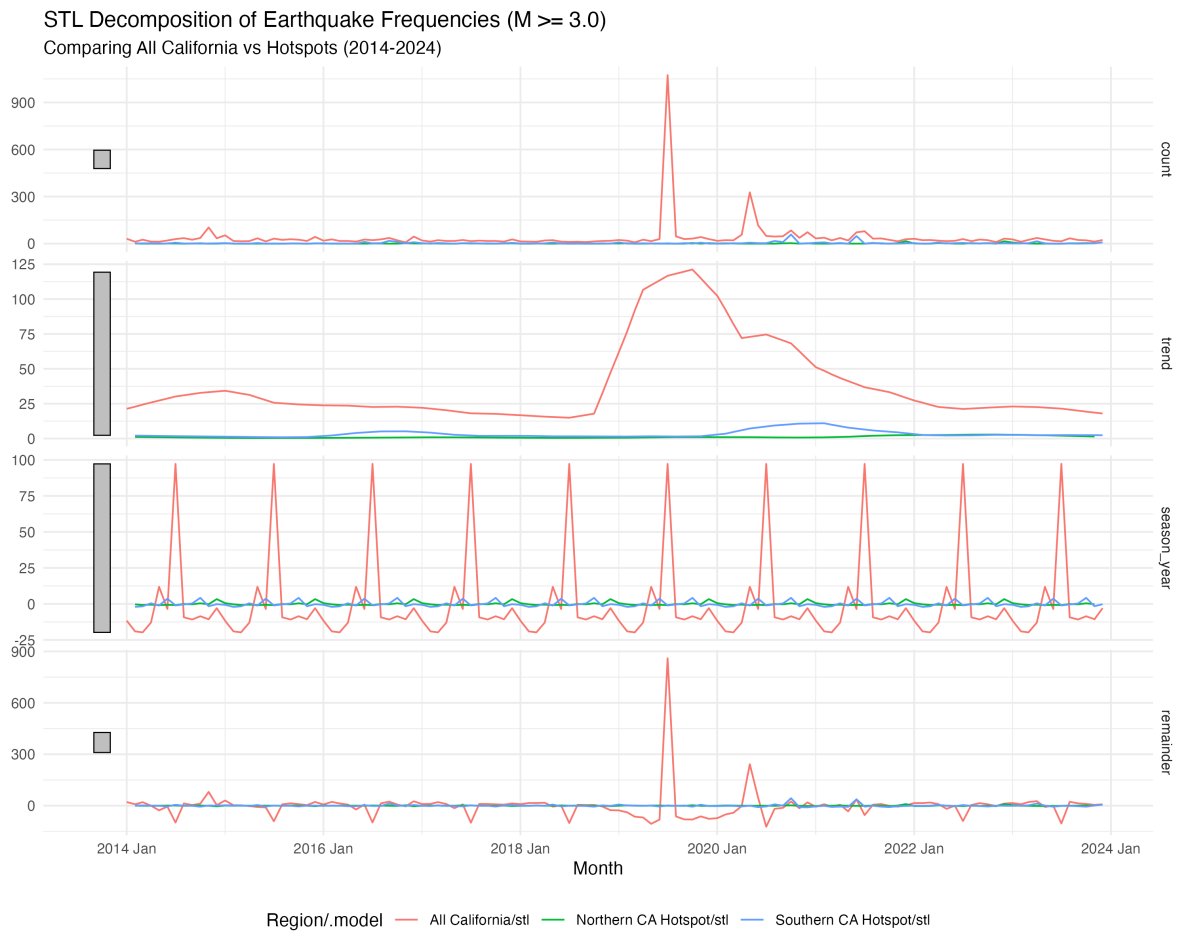
2. Time Domain Analysis

The Time Domain inherently looks backward, measuring current counts against previous lags (autocorrelation) and decomposing the rolling averages.

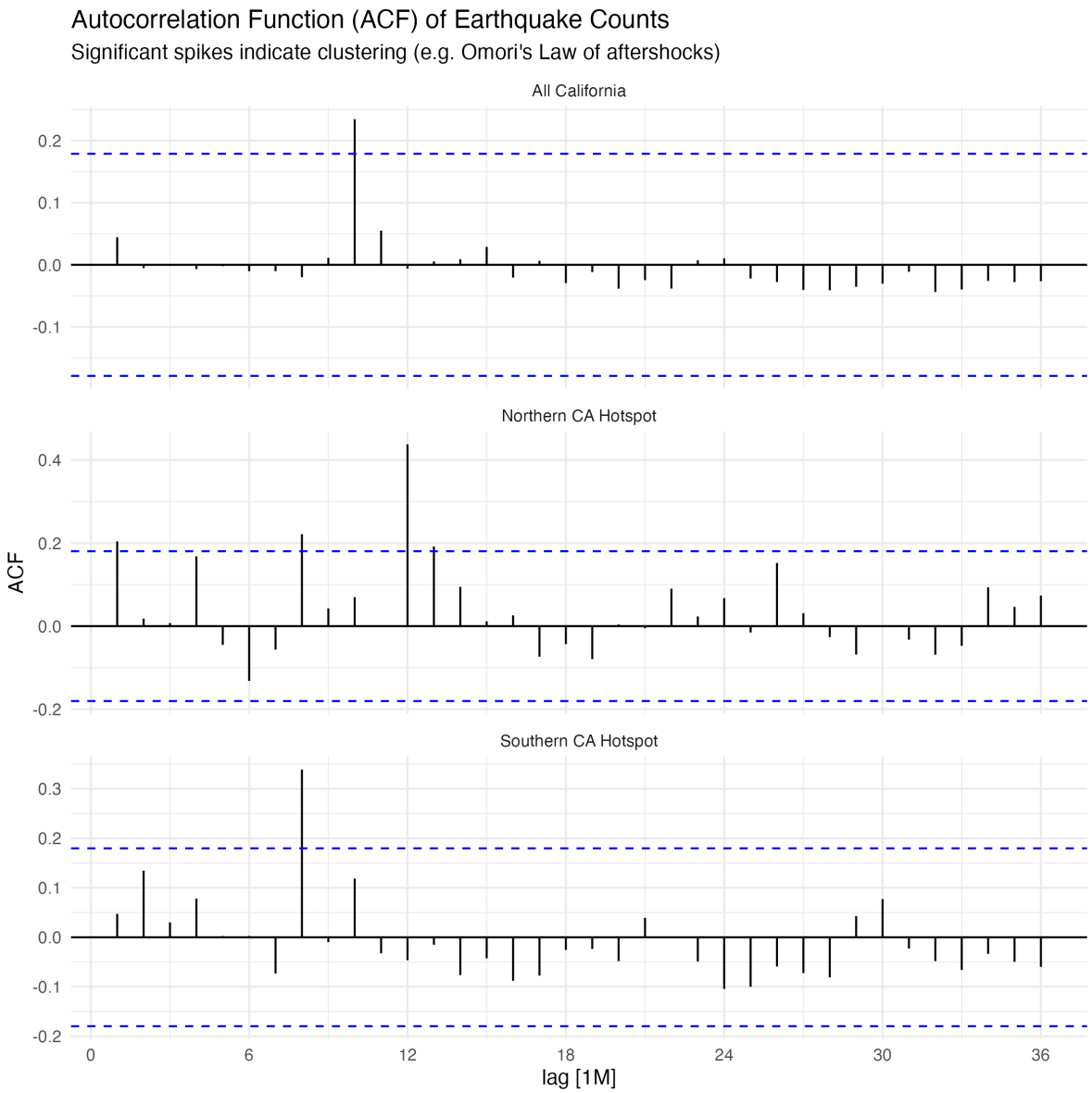
STL Decomposition



The Seasonal-Trend decomposition using LOESS (STL) reveals drastic differences across the spatial zones: * **The Northern Hotspot** exhibits a jagged, highly volatile trend with a massive noise (remainder) component, driven by distinct earthquake swarms (e.g., late 2022). * **The Southern Hotspot** is smoother, tracing a multi-year slight decline in structural activity between 2016 and 2020. * Across all regions, the scaled "Seasonality" component is statistically insignificant compared to the raw counts.

Autocorrelation (ACF)

In seismological physics, *Omori's Law* states that major quakes trigger cascading, decaying aftershocks. This physical law materializes statistically as statistically significant positive Autocorrelation.



As demonstrated, the ACF spikes reliably for the first few lags following high-rupture months across all three series, confirming the clustering nature of tectonic events.

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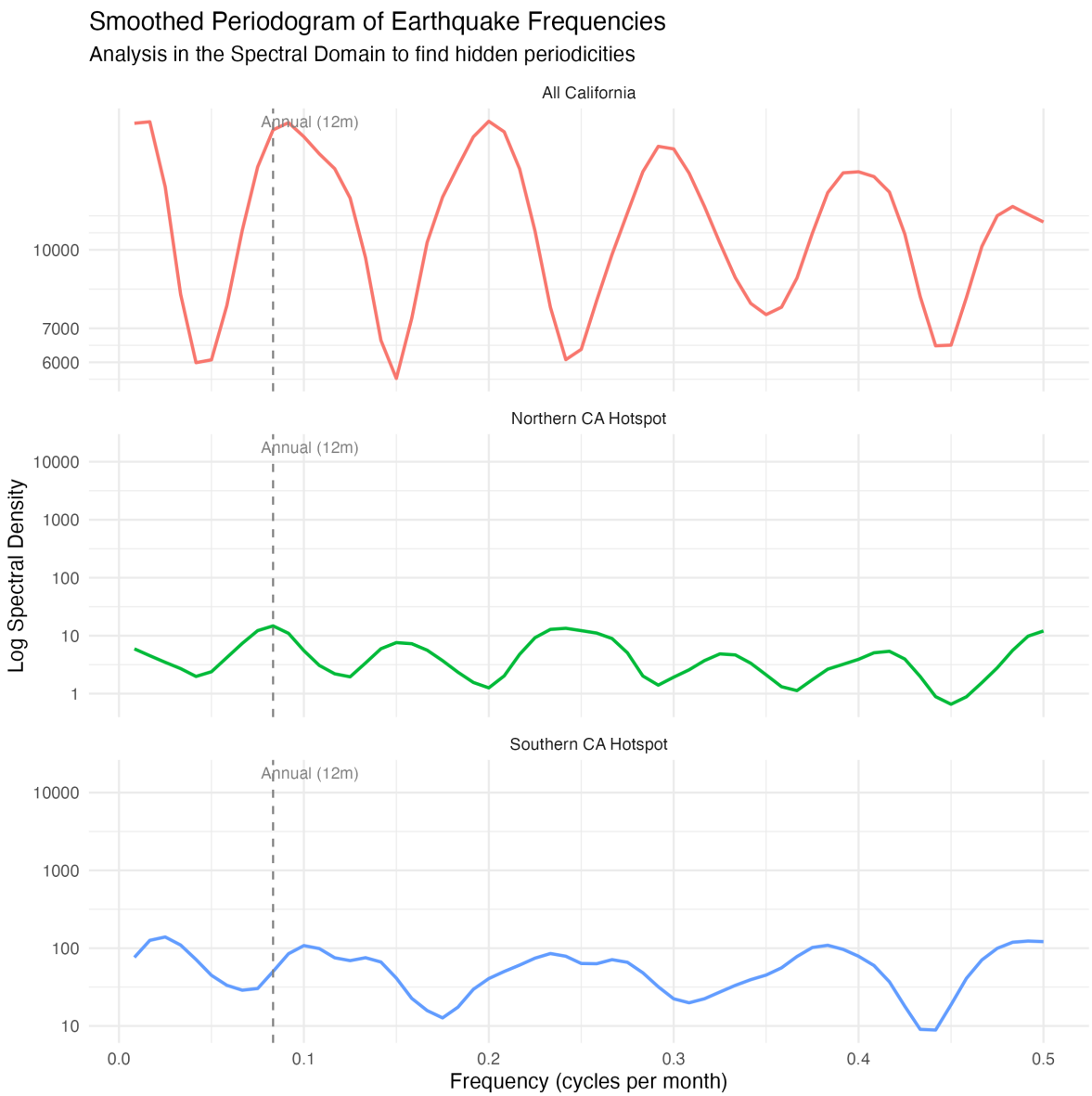
3. Spectral Domain Analysis

To rigorously test for "Earthquake Seasons" (the debated theory that events cluster around solar or lunar planetary cycles), we map the data from the Time Domain into the Spectral Domain using a smoothed periodogram.



If seismic activity correlated with the calendar year, we would expect a pronounced peak in the Spectral Density curve at fundamental frequency $1/12 \approx 0.0833$ (one cycle per 12 months). The periodograms above suggest that the density at the annual mark is negligible.

Furthermore, the spectral density peaking near Frequency 0 and slowly decaying is the textbook spectral fingerprint of an Autoregressive (AR) process, aligning perfectly with the autocorrelation findings.

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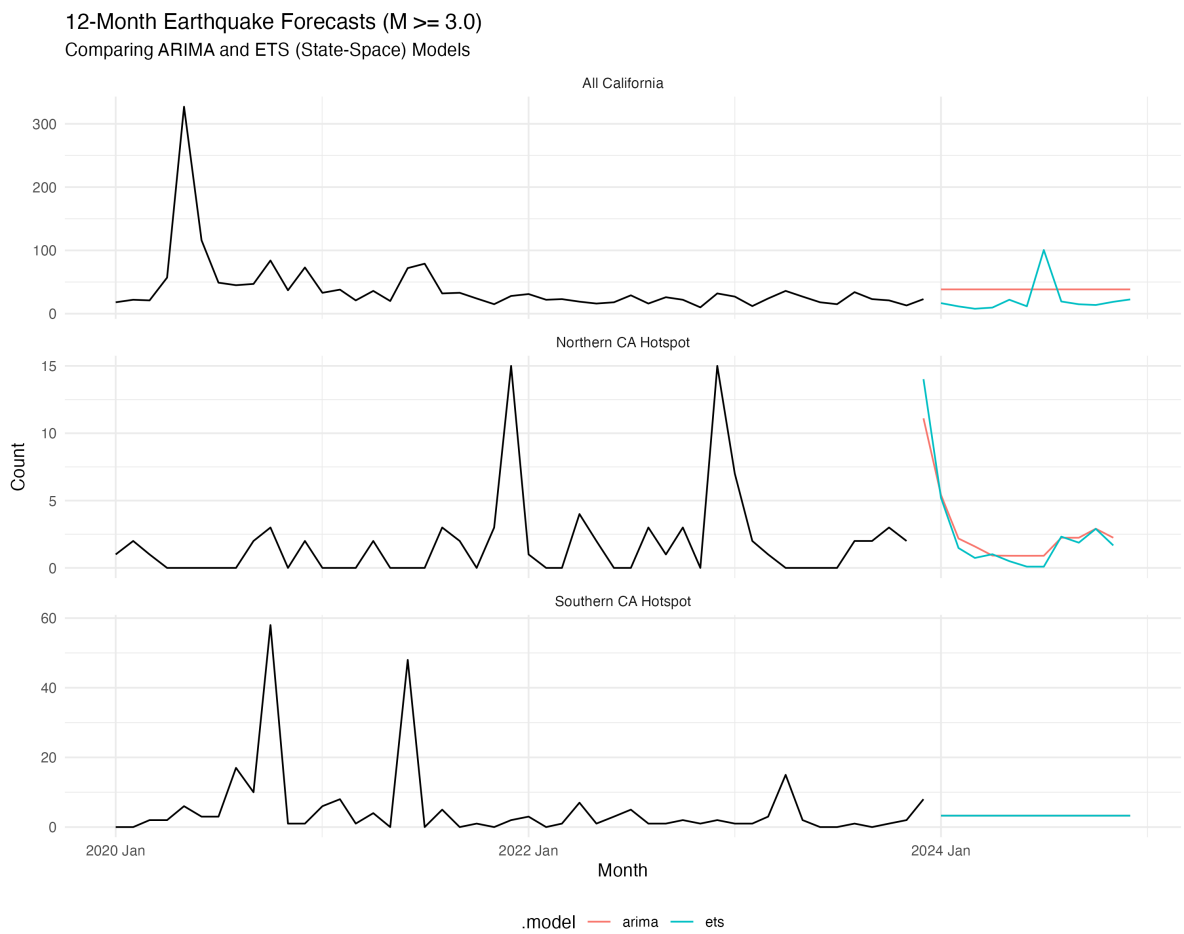
4. Modeling & Forecasting

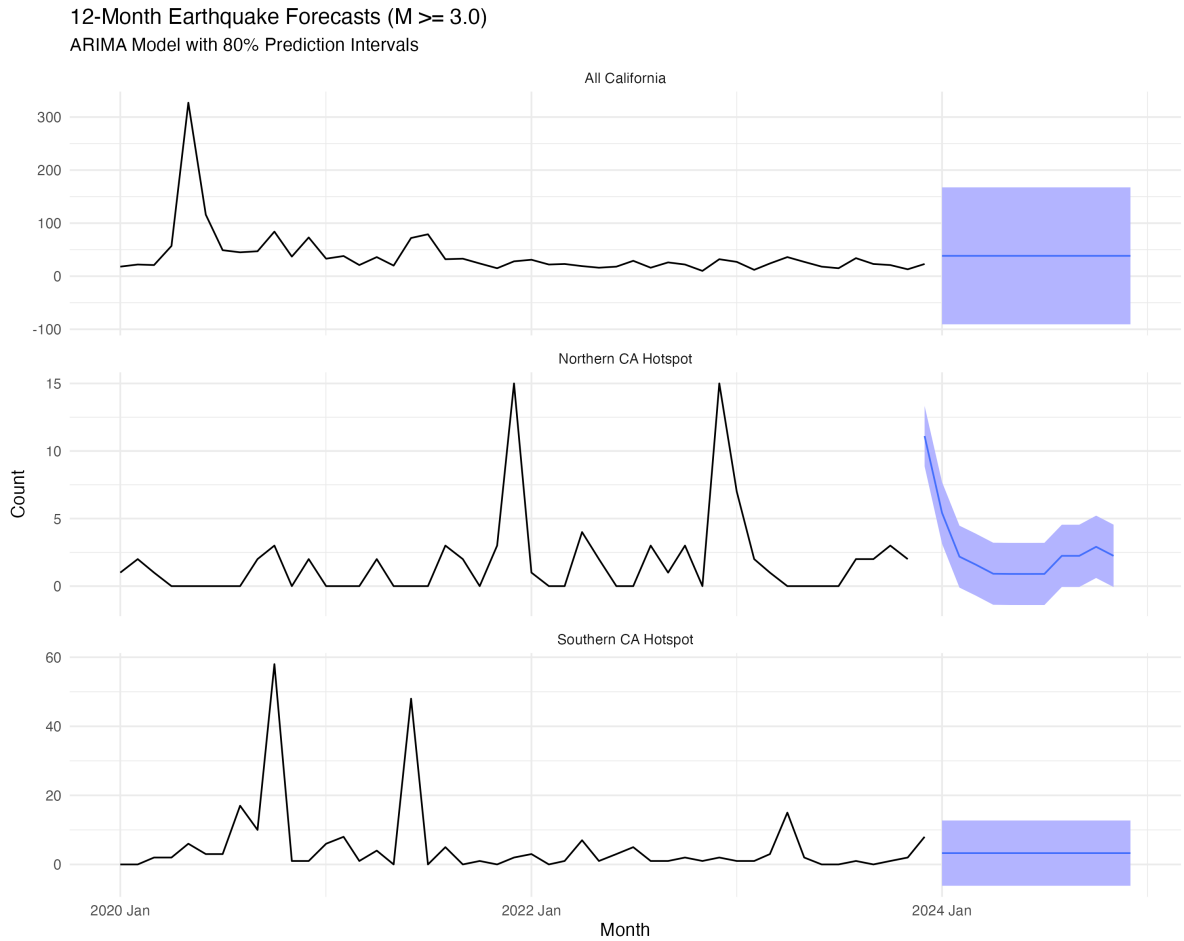
To forecast the upcoming 12 months, we fitted two distinct models minimizing the AICc information criterion:

1. **ARIMA (Auto-Regressive Integrated Moving Average):** Leveraging the strong serial autocorrelation.
2. **ETS (Exponential Smoothing / State-Space):** Treating the underlying tectonic stress as an unobserved state component.



Both models correctly anticipate the "mean-reverting" nature of earthquakes. After significant spikes, the models drag the expected values back down to the historical base rate. Due to the high variance of the Northern CA cluster sequences, prediction intervals are substantially wider than in the more predictable Southern CA region.

Below are the 80% forecast bands for the ARIMA model.



Limitations

Earthquake counts are only one summary of seismic activity. While ARIMA/ETS are useful baseline models, they do not capture the physical mechanisms of tectonic plate movement. Furthermore, data was aggregated monthly which may smooth short-term dynamics. Lastly, it may be more appropriate to fit count data using a Poisson or negative-binomial model.

Conclusion

This project demonstrates the power of fusing Geographic Information Systems (GIS) with advanced Time Series techniques. After mapping the raw events, subsetting the data spatially, and deploying models across both the Time and Spectral domains, we find little evidence for calendar seasonality while accurately modeling the autoregressive reality of earthquake cluster sequences.

Appendix: R Code Repository

The R code for this project is entirely reproducible and modularized into four main scripts: 1. `download_data.R` (API pulls & `sf` object creation) 2. `generate_maps.R` (KDE spatial heatmaps using `ggplot2`) 3. `time_domain_analysis.R` (STL, ACF modeling using `fpp3`) 4. `spectral_analysis.R` (Smoothed Periodograms) 5. `modeling_and_forecasting.R` (ARIMA, State-Space forecasting)